

Report 10 Native Android Client

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Report 10 — Native Android Client & At-Sea Safety Architecture

Project: JalJeev — Agentic Marine Intelligence System

Phase: Phase 0 (Unplanned — Built Ahead of Schedule)

Focus: Native Android app (Kotlin, Jetpack Compose), foreground safety service

1. Why a Native Android Client?

The React PWA covers judges, desk users, and browser access beautifully. But the single most safety-critical feature — **continuous background monitoring of the vessel's GPS position against marine conditions** — is something a browser tab physically cannot do.

The Browser's Limitation

- A backgrounded browser tab on Android is **throttled** by the OS (CPU, network, timers)
- After ~5 minutes in the background, the tab may be **suspended entirely**
- `setTimeout`, `setInterval`, and `fetch` reliability degrades to near-zero
- A safety warning that arrives “when the user next unlocks their phone” is **not a safety warning**

The Solution: Android Foreground Service

An Android foreground service runs with a persistent notification (“JalJeev Safety Watch — Active”), which tells the OS “this work is user-visible and must not be killed.” It gets: - **Unrestricted GPS access** while the screen is off - **Network access** in the background - **Timer reliability** for periodic safety checks - **Priority push** for safety alerts

2. Technology Stack

Component	Technology
Platform	Android (min SDK 26 / Android 8.0)
Language	Kotlin
Build system	Gradle (Kotlin DSL)
UI framework	Jetpack Compose
Map engine	MapLibre Native (Android)
Network	Retrofit + OkHttp (same FastAPI backend)
Architecture	MVVM (ViewModels + StateFlow)
Foreground service	Android <code>ForegroundService</code> with persistent notification

3. Application Structure

3.1 Four Main Screens

Screen	Purpose
Ask	Natural-language query (text + voice) — same as ChatPanel
Map	MapLibre Native map with H3 risk overlay
Route	Route/passage planning interface
Watch	Safety monitor showing current position risk status

3.2 Foreground Safety Service

The background service runs a periodic loop:

```
// Simplified flow
while (isActive) {
    val location = fusedLocationProvider.lastLocation
    val check = api.quickCheck(location.latitude, location.longitude, vesselClass)

    if (check.riskLevel in listOf("HIGH", "EXTREME", "REJECTED")) {
        notificationManager.notify(ALERT_ID, buildAlertNotification(check))
        // Vibrate + sound for HIGH/EXTREME
    }

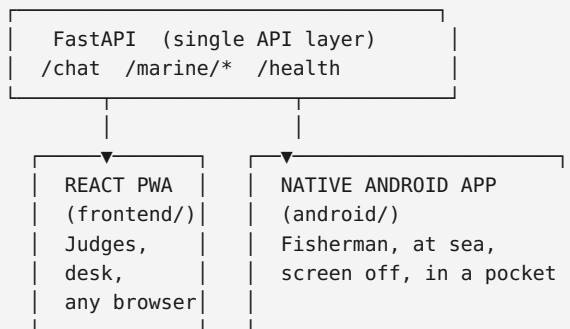
    delay(CHECK_INTERVAL_MS) // Every 5 minutes
}
```

3.3 Last-Known-Good Cache

The Android client maintains a local cache of the last successful marine state response: - If the network is unavailable (offshore), the user sees the **last known conditions** - Cache age is prominently displayed (“Data from 47 minutes ago”) - A stale cache never pretends to be current

4. API Compatibility

Both clients (React PWA and Native Android) hit the same FastAPI backend over the same HTTP surface:



Neither client holds business logic. Both render what the backend computed.

5. Build Status (as of Aug 27, 2026)

Check	Status
Gradle build clean	✓
Unit tests pass	✓
Installed on device/emulator	✗ Not yet verified
Connected to live backend	✗ Not yet verified
All tabs functional	✗ Unknown

Risk R-12

"Builds" ≠ "Runs". The Android client must be booted against the live backend and each tab screenshotted before it is shown to judges.

This is a known risk explicitly logged in the PRD.

6. Quick-Check Endpoint (/marine/quick-check)

The Watch tab relies on a lightweight endpoint designed for high-frequency polling:

- **Latency:** 0.06s warm
- **Returns:** Risk level, risk score, critical veto status, abbreviated factor list
- **No LLM call:** Pure deterministic — the watch cannot depend on an LLM
- **Cache-friendly:** Results cached per H3 cell (resolution 7, ~5 km) for 5 minutes

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